

K Prabhu

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EDUCATION

VIT-AP University, Amaravati

B.Tech in Computer Science (Specialization in Cybersecurity)

Sep 2022 – May 2026

CGPA: 7.94

EXPERIENCE

MERN Full Stack Developer — Ethnus Virtual

Jun 2024 – Aug 2024

- Built a full-stack job portal using MERN Stack with secure authentication and RESTful APIs.
- Improved system responsiveness by 30% and boosted platform engagement by 20%.
- Designed a responsive and accessible UI, increasing usability and traffic by 25%.
- Strengthened data integrity and security using MongoDB and JWT-based authentication.

PROJECTS

Deep Learning-Based Faulty Tooth Detection | *Python, YOLOv8, PyTorch, Streamlit*

- Developed an automated diagnostic system using the YOLOv8 object detection architecture to detect and localize faulty teeth in Panoramic (OPG) dental X-rays.
- Achieved a Mean Average Precision (mAP@0.5) of 86% and mAP@0.5:0.95 of 75% on the validation set by training on the Dental-Xray dataset with heavy augmentation.
- Deployed the model as an interactive Streamlit web application to visualize diagnostic results with precise bounding boxes for clinical assistance.

Network Vulnerability Scanner Tool | *Python, Socket, Threading, Matplotlib*

- Developed a Python-based vulnerability scanner to detect open ports and identify risky network services.
- Implemented multi-threaded scanning with TCP Connect Scan, reducing scan time by 50%.
- Designed automated vulnerability reporting with risk levels and mitigation strategies.
- Visualized real-time scan progress using Matplotlib.

Railway Reservation System | *MERN Stack, MySQL*

- Built a secure and scalable railway booking system using MERN stack with MySQL integration.
- Improved reservation efficiency by 40% and optimized data retrieval speed by 30%.

TECHNICAL SKILLS

- **Cybersecurity:** Kali Linux, Nmap, Wireshark, Nessus (Basic), Burp Suite (Basic), Packet Inspection, Firewall Configuration, Port Scanning, Linux Server Management, Security Auditing, Vulnerability Assessment, Ethical Hacking (Basic)
- **Languages:** Python, Java, C, SQL, JavaScript, HTML/CSS, R
- **Tools & Frameworks:** React, Node.js, Flask, Git, VS Code, MongoDB, MySQL, Google Colab

CERTIFICATIONS

- **Introduction to Cybersecurity** – Cisco Networking Academy.
- AWS Academy Cloud Architecting.
- AWS Academy Cloud Foundations.
- MERN Full Stack Certification.